

PERT /CPM Exercises

1. A factory automation project has the following activity estimates. Calculate the expected duration using the standard PERT formula. Construct the corresponding network diagram. Find out the critical path.

Activity	Description	Immediate predecessor(s)	Optimistic, <i>a</i>	Most likely, <i>m</i>	Pessimistic, <i>b</i>
A	Project planning	None	1	3	5
B	Equipment specification	A	2	5	8
C	Vendor evaluation	A	1	4	7
D	Equipment procurement	B	4	7	13
E	Site preparation	A	3	6	9
F	Installation	D, E	2	5	11
G	Staff training	C, F	2	4	8
H	Trial production	F	1	3	6
I	Final commissioning	G, H	1	2	4

2.

A product-development project has the following activities:

Activity	Duration (days)	Immediate predecessor(s)
A	5	None
B	3	None
C	6	A
D	4	A, B
E	7	B
F	5	C, D
G	3	D, E
H	2	F, G
I	4	H

1. Construct the project network.
2. Perform the forward pass to find earliest start and finish times.
3. Perform the backward pass to find latest start and finish times.
4. Calculate total float for each activity.
5. Identify all critical activities.
6. Find the planned project completion time.

3.

A project consists of the following activities with known durations:

Activity	Duration (weeks)	Immediate predecessor(s)
A	3	None
B	4	A
C	2	A
D	5	B
E	4	B, C
F	3	D, E
G	2	E
H	4	F, G

1. Draw the Activity-on-Arrow PERT network.
2. Calculate the earliest start and earliest finish times.
3. Calculate the latest start and latest finish times.
4. Determine the total float for each activity.
5. Identify the critical path and project duration.